

• H5I • PITCH

Is the Web App Your AI Built Really Secure?

The workspace to find bugs and prove correctness.

■ WEB SECURITY HAS TWO SIDES

Find vulnerabilities. Prove correctness.

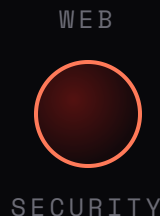
Two sides of web security.

01 · Red-teaming

Find vulnerabilities

Attack the running app the way a hacker would, and catch what nobody anticipated.

Expert hackers are expensive. Red-teaming AI can go rogue.



02 · Formal verification

Prove correctness

State the rules the logic must keep, and prove the code satisfies them.

Writing code is easy. Proving it correct is hard.

You need both, with every change the AI ships.

■ WHAT H5I DOES

Attack the running app. Prove the critical logic.

Two complementary workflows, operated by the agent through one CLI and reviewed by the developer in one UI.

01 Find vulnerabilities

ANY WEB STACK

Browser automation

Drive the app as a real user.

Traffic capture

Keep every request and response.

Replay & mutate

Change IDs, roles and inputs.

Attack patterns

Apply predefined expert techniques.

output → **finding** + **reproducible evidence**

02 Prove correctness

RUST + LEAN 4

h5i-app framework

Isolate security-critical Rust logic.

Lean extraction

Translate that logic into a proof model.

Machine proofs

Check authorization and invariants.

Verification gates

Check axioms, mutants and code drift.

output → **theorems** + **verification receipt**

One **h5i** workspace

findings

evidence

theorems

receipts

AI agent operates via CLI
Human reviews via UI

■ FIND VULNERABILITIES · ANY STACK

Give the agent the tools a security expert uses.

Browser automation, captured traffic and 1,000+ attack patterns work together through one CLI. Every result is reviewable in the UI.

AI AGENT Operates via CLI

ONE TESTING TOOLKIT

BROWSER
Browser automation
Navigate, click, fill forms and preserve authenticated state.

CAPTURE
Traffic capture
Record the exact requests and responses produced by each action.

1,000+
Attack patterns
Apply expert techniques across auth, input handling and configuration.

OPTIONAL
Supervised sandbox
Constrain files, network destinations and runtime when needed.

\$ h5i browser · h5i websec · h5i sandbox

FINDINGS + EVIDENCE



HUMAN Reviews via UI

H5I CONSOLE

HIGH f_01

Another customer's order is readable
A replay changed only the order ID. The same signed-in user received Bob's data with HTTP 200.

req_108 original browser request CAPTURED

res_109 mutated replay response CAPTURED

repro exact command and diff READY

Test what might happen. Prove what must never happen.

Build the security-critical kernel with h5i-app. Aeneas translates the Rust into Lean 4, where the rules become machine-checked theorems.

kernel/src/lib.rs · h5i-app / Rust

```
// declare the application state once
h5i_app_schema::schema! {
  mapping board_tables ...;
  pub struct Snapshot {
    posts: Vec<Post>,
    moderators: Vec<Moderator>
  }
}

// pure logic that ships
pub fn transition(actor, state, cmd)
  -> Outcome { ... }
```

AENEAS
EXTRACTS



proofs/Theorems.lean · Lean 4

```
import Commands

/-- Every successful write is allowed. -/
theorem authorized
  (h : transition a s c = .ok result) :
  ∀ w ∈ writes result,
    allowed s a.user w := by
  ...

theorem inv_preserved ... := by
  ...
```

LEAN
CHECKS



GUARANTEES

authorized

Every successful write is allowed by policy.

invariant preserved

Every reachable state keeps the rules.

transition total

No panic, overflow or invalid index.

■ WHY H5I

Other tools give you one side. h5i brings both together.

Browser + proxy tools

Built for people. The agent juggles browser, proxy, and CA setup. No proofs.

Sandboxes

Limit what the agent can reach. No testing, no evidence, no proofs.

Proof assistants

Prove a model of the logic. Nothing connects it to the running app.

h5i

Red-team the running app and prove the logic, in one CLI the AI runs and one console you review.

■ TRACTION & USERS

Built in the open.

Built for teams that ship with AI.

670+

GitHub stars

60+

forks

20+

contributors

2.5k+

release downloads

GitHub · checked October 2, 2026 · [Repository and releases](#)

Product teams

shipping with AI, no security specialist

Security engineers

testing between assessments, with evidence

Rust teams

with rules that must never break

■ NEXT

Find vulnerabilities. Prove correctness. For every team that ships with AI.

Product

findings beside the proofs they touch, in one console

Adoption

10 design partners, then the first paid pilots

Open source

local-first, Apache 2.0, built with the community

Try it with your own AI agent: `curl -fsSL https://h5i.dev/install.sh | sh -s -- --websec --test`

h5i.dev

github.com/h5i-dev/h5i